# IRexp experimental infrared band lists from the PMC Open Access Subset and Chemotion

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## Abstract

IRexp is a redistributable collection of **experimental infrared band lists** — author-reported peak positions in cm $^{1}$ — mined from open chemistry literature, optionally paired with $^{1}$H/$^{13}$C shift lists and resolved structures. The release holds **121,233** records (119,345 from the PubMed Central Open Access Subset; 1,888 from Chemotion/RADAR4Chem), of which **43,060** are structure-linked and **33,201** are full IR + $^{1}$H + $^{13}$C + structure quadruples. IRexp stores **numeric band lists**, not absorbance traces or JCAMP-DX. Records carry `source_doi` for attribution. Licensing is **mixed**: Chemotion rows are CC-BY-SA-4.0; PMC Open Access uses diverse Creative Commons terms, with per-article commercial/non-commercial segregation pending ([PENDING: commercial count after licence join]). Intended reuse includes multimodal spectral model training, literature-scale cheminformatics, and complementary elucidation benchmarks that cite this Descriptor. Data are on Hugging Face (`ilkhamfy/IRexp`); a Zenodo DOI will follow licence remediation ([TODO: 10.5281/zenodo.XXXXXXX]). Harvest, extraction, and validation code is MIT-licensed.

## Background & Summary

Infrared (IR) spectroscopy is routine in organic characterisation, yet **open, redistributable** collections of *experimental* IR data remain sparse relative to modern machine-learning needs. Digitised absorbance libraries such as the NIST Chemistry WebBook[1] and AIST SDBS[2] are valuable but either modest in size or **view-only** (no bulk redistribution). Computational IR–NMR resources published in *Scientific Data* expand multimodal coverage with simulated spectra[3], while large literature mines for **NMR** peak lists (notably NMRexp[4]) demonstrate that peer-reviewed experimental spectral *lists* with DOI traceability are in scope for this journal. Concurrent literature corpora that include IR among other modalities (for example NMRSpec/NMRTrans[5]) further motivate an **IR-focused, redistributable** band-list resource rather than another closed spectrum archive.

IRexp fills a specific wedge. Experimental sections of chemistry papers conventionally report per-compound **IR band lists** (wavenumbers in cm $^{1}$) together with $^{1}$H/$^{13}$C NMR shift lists. That textual convention is the object language models and many elucidation pipelines consume, and it is a **different object** from a digitised spectrum. IRexp therefore:

1. Harvests open-access full text from the **PMC Open Access Subset** bulk S3 mirror and peak lists from the **Chemotion** FT-IR deposit.
2. Extracts deterministic numeric IR band lists (and co-reported NMR strings where present).
3. Resolves compound names to canonical structures with OPSIN[6], RDKit[7], and SELFIES[8] where possible.

4. Releases **extracted numbers only** — no PDFs, figures, or article full text — with source accessions for attribution.

Among openly redistributable *text-derived IR band lists*, IRexp is large by record count (121,233). It does not claim to replace SDBS or NIST absorbance libraries; SDBS alone holds more structure-linked *spectra* than IRexp holds structure-linked band lists. The intended reuse is multimodal pre-training, supervised IR→structure modelling on `irexp_resolved`, and as the literature substrate for complementary elucidation benchmarks described elsewhere (forthcoming ICLR manuscript on IRSpectra-Bench; cite that work for protocol and model results — **not** reproduced here).

**What this Data Descriptor does not contain.** No hypothesis tests, no large-language-model accuracy tables, and no stage-decomposition of elucidation performance. Those belong in the complementary research paper.

## Methods

### Source corpora

**PMC Open Access Subset.** Primary harvest uses the NCBI PMC OA bulk distribution on Amazon S3 (`s3://pmc-oa-opendata`, HTTPS endpoint `https://pmc-oa-opendata.s3.amazonaws.com`)[9]. A harvest snapshot records **188,016** distinct PMC identifiers scanned (`data/irexp/seen_papers.txt.gz`). The released corpus retains records from **15,416** unique `PMC:*` accessions. Extraction operates on open-access plain text / JATS available through that OA path (and Europe PMC full-text XML for validation re-fetches). The **released** IRexp DOIs are exclusively `PMC:*` accessions or the Chemotion deposit DOI; no paywalled publisher DOI appears in the 121,233-record file.

PMC OA is **not** a single licence. The subset is partitioned into commercial-use, non-commercial, and other packages[9]. A Europe PMC licence sample of 200 unique PMCIDs drawn from IRexp found approximately 73% `cc by`, approximately 19% NC or empty licence fields, and the remainder other/empty/no-result (`data/NOTICE`). **Per-article licence join and segregation of commercial vs non-commercial pools is Track 1 work and is not yet complete** — see Data Records placeholders.

**Chemotion / RADAR4Chem. 1,888** records come from the Chemotion Repository FT-IR collection deposited at RADAR4Chem (DOI `10.22000/OGoEQGlsZGElrgst`)[10], licensed **CC-BY-SA-4.0**. These rows are **peak lists derived from deposited experimental FT-IR spectra** (not author-transcribed experimental-section prose). They carry a stamped `license=CC-BY-SA-4.0` and `source_doi` equal to the deposit DOI. Median band count is **39** versus **9** for PMC-transcribed lists; users modelling band density should treat the pools separately.

**Excluded.** AIST SDBS (view-only; no bulk export)[2]; NIST WebBook join code exists in the repository but contributes **0** records to the released IRexp (`ir_source` is always `experimental` for released rows).

### Discovery and fetch (released corpus)

1. Enumerate PMC OA identifiers / packages from the S3 OA mirror.
2. Fetch OA full text for experimental-section mining.
3. Ingest Chemotion deposit peak lists with structure metadata from the RADAR4Chem release.
4. Persist harvest provenance in `seen_papers.txt.gz` and optional rawer rows in `ir_harvest_snapshot.jsonl`

(134,893 rows including intermediate prose fields used to diagnose materials-text false positives; the curated release is `irexp.jsonl.gz`).

Development adapters for additional publishers exist in the codebase for exploration; they are **outside the construction path of the released dataset**. Methods for IRexp as published here are restricted to **PMC-OA S3 + Chemotion**.

### Extraction (band lists, not spectra)

A deterministic regular-expression pipeline (`spectro_scraper/extract.py`) segments experimental text per compound and extracts:

- IR wavenumbers → `ir_bands_cm-1` (list of floats, cm $^{1}$);
- $^{1}$H and $^{13}$C payloads → `h_nmr` / `c_nmr` (author strings, when present).

Gates reject scan-range artefacts and common prose false positives (for example hydrogel / materials narrative mistaken for IR lists). **No curve digitisation** is performed: figures are not traced; JCAMP-DX is not stored.

### Structure resolution

Where an IUPAC or systematic name is available, names are converted with OPSIN (py2opsin)[6], canonicalised with RDKit[7], and encoded as InChIKey and SELFIES[8]. An optional PubChem[11] name fallback (`USE_PUBCHEM`) handles trivial names. Structure coverage of the full release is **35.5%** (43,060 / 121,233). The structure-complete split is shipped as `irexp_resolved`.

### Licence handling (current vs planned)

| Pool | Records | Stamp today | Planned (Track 1) |
|---|---:|---|---|
| Chemotion | 1,888 | `CC-BY-SA-4.0` | unchanged |
| PMC | 119,345 | `license` unset / null on disk | join each PMCID → Europe PMC / PMC OA package licence; segregate commercial / NC / other; stamp every row |

`scripts/split_license_pools.py` separates Chemotion vs PMC **by DOI prefix** for physical pool files. Until Track 1 lands, treat its PMC output as "PMC-sourced", **not** as a verified CC-BY-4.0 commercial pool (`data/NOTICE`).

### Quality tooling

Optional physics gates live in `spectro_scraper/quality.py` ($^{13}$C peak count   carbon count; $^{1}$H integration vs formula; IR wavenumber windows). They were **not** applied as a hard filter to every released row at harvest; Technical Validation reports post-hoc checks below. Transcription fidelity uses `scripts/audit_extraction.py`.

## Data Records

### Object definition

Each IRexp record is a JSON object. Required chemistry fields for an IR entry:

| Field | Type | Description |
|---|---|---|
| id | string | Stable record id (InChIKey when resolved, else internal) |
| ir_bands_cm-1 | float list | Experimental IR peak positions (cm $^{1}$) |
| ir_source | string | experimental for all released rows |
| source_doi | string | PMC:<pmcid> or Chemotion deposit DOI |
| h_nmr / c_nmr | string or null | Author-reported shift lists |
| smiles / selfies / inchikey | string or null | Resolved structure encodings |
| has_structure | bool | Convenience flag |
| license | string or null | Stamped for Chemotion; **PMC pending Track 1** |

**Not included:** absorbance traces, intensities, instrument metadata beyond what appears in source text, PDFs, figures, or full article bodies.

### Files and counts

Paths relative to the project repository / Hugging Face mirror.

| File | Records | Description |
|---|---|---|
| data/irexp/irexp.jsonl.gz | 121,233 | Full curated release |
| data/irexp_resolved/irexp_resolved.jsonl.gz | 43,060 | 100% structure-linked |
| data/irexp_release/train_no_bench.jsonl.gz | 42,808 | Resolved minus IRSpectra-Bench InChIKeys |
| data/irexp_release/train_no_bench_nmr.jsonl.gz | 32,949 | Same with both $^{1}$H and $^{13}$C |
| data/irexp_release/pretrain_ir.jsonl.gz | 119,345 | PMC-only IR pretrain pool |
| data/irexp/seen_papers.txt.gz | 188,016 lines | PMC IDs scanned at harvest |
| data/irexp/ir_harvest_snapshot.jsonl.gz | 134,893 | Intermediate harvest snapshot |

**Composition of irexp.jsonl.gz:**

| Slice | Count |
|---|---|
| All IR band-list records | 121,233 |
| With $^{1}$H and/or $^{13}$C NMR | 87,075 (72%) |

| Slice | Count |
|---|---|
| With resolved structure | 43,060 (35.5%) |
| Structure + any NMR | 40,702 |
| Full IR + $^{1}$H + $^{13}$C + structure | 33,201 |
| PMC OA provenance | 119,345 |
| Chemotion provenance | 1,888 |
| Unique PMC accessions | 15,416 |

**Licence-pool counts after Track 1** (do not invent):

| Pool | Count |
|---|---|
| PMC commercial-use (CC BY / CC0 / …) | `[PENDING: commercial count after licence join]` |
| PMC non-commercial (CC BY-NC*) | `[PENDING: NC count after licence join]` |
| PMC other / empty / unresolved | `[PENDING: other count after licence join]` |
| Chemotion CC-BY-SA-4.0 | 1,888 |

Median bands: **9** (PMC), **39** (Chemotion). All **1,360,866** released IR band values fall inside 350–4000 cm $^{1}$ (full-corpus range check; Technical Validation).

### Access

- **Hugging Face:** https://huggingface.co/datasets/ilkhamfy/IRexp (public mirror; re-issue after Track 1).
- **GitHub development mirror:** https://github.com/IlkhamFY/spectro-agent
- **Zenodo archival snapshot:** `[TODO: 10.5281/zenodo.XXXXXXX]` (mint after honest multi-licence metadata).

## Technical Validation

### Transcription fidelity (PMC)

On a seed-fixed random sample of **60** PMC-sourced records (`scripts/audit_extraction.py --n 60 --seed 0`), each article was re-fetched from Europe PMC and every recorded wavenumber was checked against the source text (±1 cm $^{1}$ integer match). Result: **560/560 bands** and **60/60 records** confirmed (Wilson 95% CI  99.3–100% for bands;  94–100% for records). Frozen machine-readable summary: `data/audit/extraction_audit.json`.

This bounds **transcription** error (hallucinated, mis-parsed, or unit-mangled values). It does **not** measure extraction **recall** (whether every IR string in every paper was found) — that requires human reading of full experimental sections and remains planned (see below).

### Structure–NMR consistency (resolved sample)

On a random sample of **500** `irexp_resolved` records with $^{13}$C text (seed 0), comma-separated $^{13}$C peaks were counted against RDKit carbon counts: **17/500 (3.4%)** listed more peaks than carbons (physically impossible → name/structure assignment or parse error). On **500** records with $^{1}$H text, summed reported integrals exceeded formula hydrogens by more than 2 in **17/497** scored rows (**3.4%**; 3 lacked parseable integrals). Reproducible summary: `docs/scientific_data/qc_structure_nmr.json`.

These rates are reported as integrity diagnostics for re-users (quarantine or filter before supervised training). They are **not** claimed as a full expert structure audit; NMRexp-scale manual skeleton checks (n 300) remain future work.

### IR physical window

Every band in the full 121,233-record release lies in **[350, 4000] cm** $^{1}$ (0 / 1,360,866 out of range).

### Planned / deferred QC

| Check | Status |
|---|---|
| Per-PMCID licence join + pool counts | **Pending Track 1** |
| Extraction-recall human audit (papers, not bands) | Planned (n 30–50 papers) |
| Larger transcription sample (n 200) | Planned |
| Full-corpus `quality.py` quarantine pass | Planned |
| Expert structure spot-check (n 100) | Optional / deferred |

## Usage Notes

- **Band lists  spectra.** Do not evaluate models trained on IRexp as if they had seen full absorbance curves.
- **Separate pools by density and licence.** PMC (sparse, author-transcribed) vs Chemotion (denser, deposit peak-picked). Keep Chemotion ShareAlike constraints in mind when combining pools; combined redistribution of Chemotion-derived rows must honour CC-BY-SA-4.0.
- **Do not assume PMC = CC-BY-4.0.** Wait for Track 1 stamps or restrict use to research compatible with mixed upstream terms and per-article attribution via `source_doi`.
- **Training without benchmark leakage.** If using the complementary IRSpectra-Bench problems, fine-tune from `train_no_bench.jsonl.gz` (or rebuild with `scripts/build_train_no_bench.py`) so benchmark InChIKeys are withheld.
- **Structure coverage.** Prefer `irexp_resolved` for supervised structure tasks; 64.5% of records lack SMILES.
- **Attribution.** Cite this Data Descriptor / Zenodo DOI and attribute originating articles through each record's `source_doi`.

## Data Availability

IRexp numeric extracts are available at:

- Hugging Face Datasets: https://huggingface.co/datasets/ilkhamfy/IRexp
- GitHub: https://github.com/IlkhamFY/spectro-agent (`data/irexp/`, `data/irexp_resolved/`, `data/irexp_release/`)

- Zenodo: `[TODO: 10.5281/zenodo.XXXXXXX]` (archival DOI at proof, after Track 1 licence metadata)

Licensing summary (honest):

- **Chemotion (1,888):** CC-BY-SA-4.0[10].
- **PMC (119,345):** mixed Creative Commons / PMC OA package terms — **not** uniformly CC-BY; `[PENDING: commercial count after licence join]`.
- Only extracted numeric fields and identifiers are redistributed; source full texts are not.

## Code Availability

Harvesting, extraction, structure resolution, licence pool splitting, and validation scripts are in https://github.com/IlkhamFY/spectro-agent under the **MIT License** (`LICENSE`). Principal modules: `spectro_scraper/` (`extract.py`, `normalize.py`, `pipeline.py`, `quality.py`); validation `scripts/audit_extraction.py`; pool split `scripts/split_license_pools.py`. The version corresponding to this Descriptor will be tagged at Zenodo deposit time.

## Author contributions

**I.Y.:** conceptualization, methodology, software, data curation, validation, writing — original draft.
**R.S.:** methodology, software, validation, writing — review and editing.
**R.A.V.-H.:** conceptualization, supervision, writing — review and editing.

## Competing interests

The authors declare no competing interests.

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## References

References are maintained in `docs/scientific_data/references.bib` (pandoc/CSL build) and overlap the project bibliography where shared.

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